PROPOSAL

CONFIDENTIAL




EXECUTIVE SOLUTION & TECHNICAL ARCHITECTURE PROPOSAL

# Parcel Delivery Platform

Product Vision | System Architecture | Technology Strategy | Implementation Roadmap

PREPARED FOR

[Client Name]
[Client Organisation]

PREPARED BY

[Author Name]
[Company]

DATE AND VERSION

22 October 2026
Version 1.0 — for discussion

DOCUMENT INFORMATION

| | |
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| Prepared by | [Author Name], [Company] |
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## How to read this document

The proposal separates what the client has asked for, what the development team recommends, what already works, and what is still open. Every material statement about the product carries one of the labels below, so that no recommendation is mistaken for an agreed rule, and no plan is mistaken for finished work.

| LABEL | MEANING |
|---|---|
| CLIENT BRIEF | Stated by the client. The brief so far is short: a delivery application in the spirit of inDrive, dedicated to parcels, with a request for the product vision, architecture and development approach. |
| BUILT & TESTED | Implemented in the prototype and exercised by automated tests that run against a real database. Not yet proven in the field with real users, devices and networks. |
| PARTLY BUILT | Implemented in part, or implemented but not yet connected to the real external service it depends on. |
| RECOMMENDED | An engineering or product recommendation from the development team, with its reasoning. Open to discussion. |
| CLIENT DECISION | A business, commercial or legal choice that only the client can make. Shown with a recommendation where one is appropriate. |
| TO BE VERIFIED | A fact that could not be confirmed from an authoritative source at the time of writing, such as a price or a legal interpretation. |

Prices are quoted in US dollars as published by the providers, before tax, and only where a source was checked; see Appendix A for each source and its verification status. Dinar amounts shown in screenshots come from test configuration values, not from proposed tariffs.

SECTION 1

# Executive overview

The client asked for an application that works like inDrive but carries only parcels. We propose a city delivery marketplace in which the sender sees a fair recommended price, the nearest suitable courier accepts the job within seconds, and the parcel, the price and any cash collected are protected from the moment of acceptance to the moment of delivery.

## 1.1 Our understanding of the idea

inDrive is known for letting passengers propose their own fare and letting drivers respond. Applied to parcels, the useful part of that idea is price transparency and customer choice. The less useful part is open-ended negotiation, which slows matching and invites disputes after the fact. A parcel service also carries risks that ride-hailing does not: the goods travel without their owner, the recipient is often a third party, and in Algeria many deliveries involve cash on delivery collected on behalf of online sellers.

Our reading of the brief is therefore a dedicated parcel marketplace for individuals and online sellers, launched in Algiers and designed to extend to other cities without being rebuilt. CLIENT BRIEF covers the inDrive reference and the parcel-only focus. Algiers as the launch city and the operating rules described in this document are planning assumptions and recommendations that the client is asked to confirm.

## 1.2 Proposed vision

To be the most trusted way to send a parcel across an Algerian city: quick to order, priced fairly and predictably, and safe for the goods and the money they carry. Trust, rather than breadth of services, is the intended differentiator.

## 1.3 The solution at a glance

| CHARACTERISTIC | WHAT IT MEANS FOR USERS | STATUS |
|---|---|---|
| **One app, two journeys** | The same Android and iPhone app serves senders and couriers; one phone number can hold both profiles, kept separate. | BUILT & TESTED |
| **Recommended price, sender in control** | The platform recommends a price from road distance and parcel size; the sender may adjust it within limits set by operations, and raise it while waiting. | BUILT & TESTED<br>CLIENT DECISION |
| **Price fixed at acceptance** | No renegotiation once a courier accepts. | BUILT & TESTED |
| **Automatic matching** | The job is offered to the nearest eligible courier first, one at a time, instead of an open auction. | BUILT & TESTED |
| **Proof of delivery** | A four-digit code held by the sender and given to the recipient completes the delivery. | BUILT & TESTED |
| **Cash-on-delivery accountability** | Every amount a courier collects is recorded in an append-only ledger, with per-courier limits and numbered handover receipts. | BUILT & TESTED |
| **Operations console** | Courier approval, live orders, interventions, problems, cash, zones and prices, with a permanent audit trail. | BUILT & TESTED |

| CHARACTERISTIC | WHAT IT MEANS FOR USERS | STATUS |
|---|---|---|
| **Google Maps** | Maps, address search and travel times from Google Maps Platform in production. | RECOMMENDED |
| **Arabic, French, English** | Full right-to-left Arabic and instant language switching, including notifications. | BUILT & TESTED |

## 1.4 Where development stands

The project is past the concept stage. A working prototype of all three applications exists and its automated test suite passes in full. It has not yet been deployed to a public server or used by real couriers, and several production integrations remain to be connected.

| | | | |
|---|---|---|---|
| **3** applications working together: mobile app, operations console, backend | **56** server functions behind the app and console | **358** automated tests passing, 92 of them attack scenarios (run 22 Oct 2026) | **12** database integrity rules verified by direct checks |

An Android test build of the app has been produced through Expo's cloud build service. It is not yet connected to a reachable server, so it cannot place real orders today. The prototype's maps use open-source components; the production plan replaces them with Google Maps (Section 5.2). The SMS login provider, phone push notifications and the hosting environment are designed and partly built but not yet connected to live accounts. Section 3 lists exactly what is complete, simulated and outstanding.

## 1.5 Recommended architecture

We recommend keeping the current architecture, which is deliberately simple and already tested: a single backend application organised into clear business modules (a *modular monolith*), a PostgreSQL database that enforces the money-related rules itself, a React Native mobile app shared between Android and iPhone, and a web console for operations. For hosting we recommend Google Cloud managed services in a lean configuration for the pilot, Cloud Run with Cloud SQL, adding high availability, a shared cache and a web application firewall before public launch. Kubernetes and microservices are not justified at this stage; the design allows either later if load requires it.

## 1.6 Path to production

We propose five phases, each closed by explicit completion criteria rather than fixed dates: business decisions and accounts; production engineering; verification and security testing; a closed pilot in selected Algiers zones; and public launch, Android first. The decisions that most affect scope and timing are the commercial model (commission, courier payment, fees), the hosting location under Algerian data-protection law, and the choice of SMS provider. Sections 6 and 7 set out the plan, the costs that can be estimated today, the main risks and our recommended next steps.

SECTION 2

# Product and operating model

Four parties take part in every delivery: the sender, the courier, the recipient and the operations team. The platform coordinates them and holds the rules that protect each of them: the agreed price, the parcel, the cash, and the record of what happened.

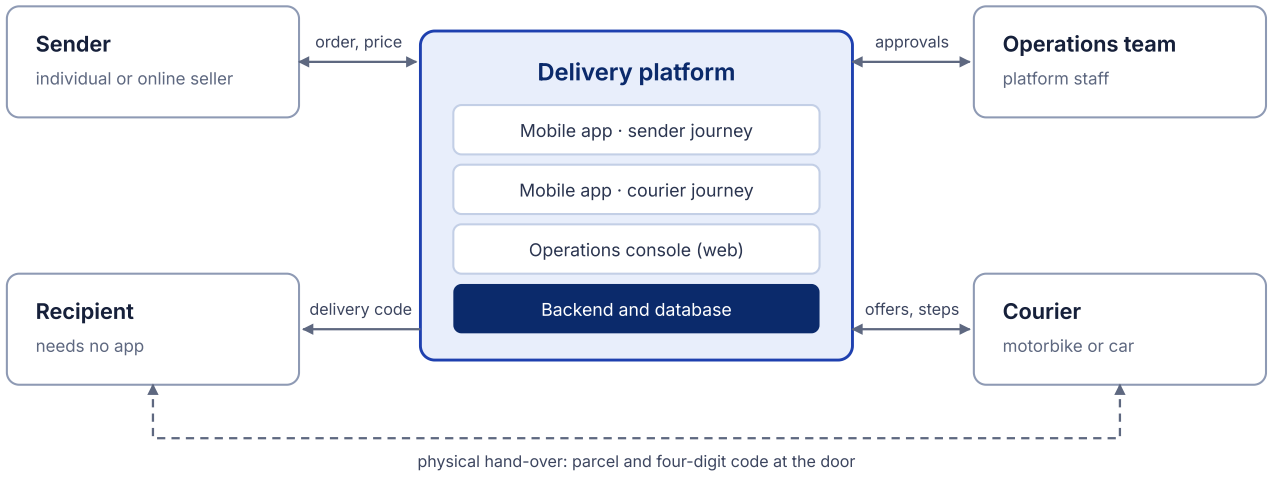



**Figure 1 — Platform ecosystem.** Solid lines are digital interactions through the platform; the dashed line is the physical hand-over that the delivery code verifies. The courier also collects the parcel from the sender at the pick-up address.

## 2.1 Roles and experiences

| ROLE | EXPERIENCE | STATUS |
|---|---|---|
| **Sender** | Logs in with an SMS code; sets pick-up and drop-off on a map with street, building, floor and landmark; enters parcel size, recipient and any cash-on-delivery amount; adjusts the price; follows the courier; shares the delivery code; rates; reports problems. | BUILT & TESTED |
| **Courier** | Signs up with identity, licence and vehicle photos; once approved, goes online and receives offers showing price, distances, cash to collect and sender rating; follows one action at a time; confirms delivery with the code and cash collected; records failed deliveries, which start a return. | BUILT & TESTED |
| **Recipient** | Needs nothing installed. If the recipient's number belongs to a registered user, the app shows the incoming parcel and its code; otherwise the sender passes the code on. | BUILT & TESTED |
| **Operations** | Separate staff accounts with two-factor login. Approve couriers and set their cash limits; follow and intervene in orders; resolve problems; record cash handovers; manage service zones and prices. Every action is logged permanently. | BUILT & TESTED |

## 2.2 Delivery lifecycle

An order moves through a small set of states, and only the transitions shown below are possible. The database itself rejects any other change, so a fault in the application cannot, for example, mark an order delivered before it was picked up.

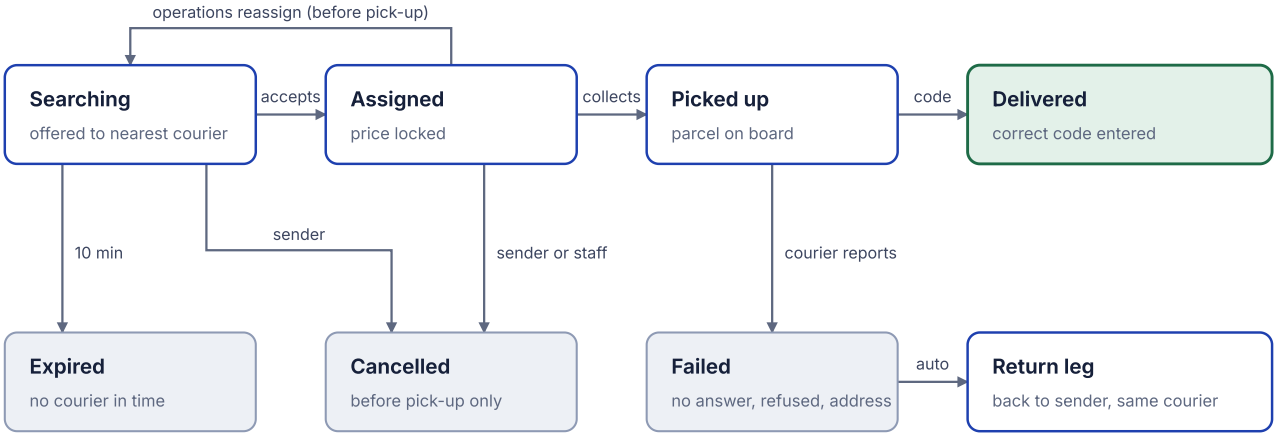



**Figure 2 — Order lifecycle.** Final states: delivered, cancelled, expired, failed. A failure creates a return leg and a problem report; five wrong codes lock the order for staff review.

## 2.3 Pricing: inDrive-style negotiation or a guided price?

The inDrive reference does not by itself settle how prices should work, so we set out the choice explicitly. In a negotiated model the customer proposes a fare and drivers may counter-offer. In the model built in the prototype, the platform recommends a price, the sender may move it within a corridor and raise it while waiting, and couriers simply accept or decline.

| | OPEN NEGOTIATION (INDRIVE-STYLE) | GUIDED PRICE (BUILT IN PROTOTYPE) |
|---|---|---|
| **Customer control** | High; any offer, counter-offers possible | High within a corridor; can raise at any time while searching |
| **Speed of matching** | Slower; several rounds and parallel offers | Faster; one offer at a time to the nearest courier, 20 seconds each |
| **Price predictability** | Varies widely for the same trip | Consistent, derived from road distance and size |
| **Abuse and disputes** | Risk of very low offers and renegotiation at pick-up | Floor and ceiling set by operations; price locked at acceptance |

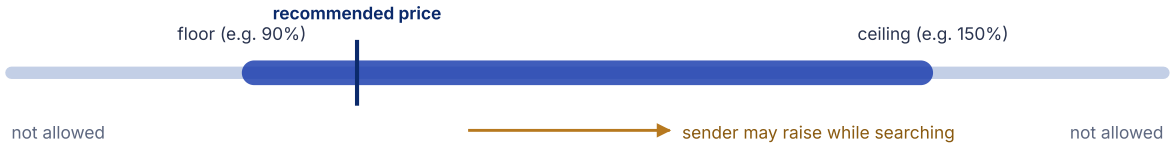



**Figure 3 — Price corridor.** Recommended price = base fee + rate per kilometre of road distance + size and vehicle supplements, rounded up to 10 DA. Operations set every value, and the corridor percentages, from the console; each order keeps the price version it was created with. The percentages shown are examples, not proposed tariffs.

RECOMMENDED Keep the guided price. It preserves the customer choice that makes inDrive attractive while avoiding its best-known weakness, renegotiation after acceptance. CLIENT DECISION The client should confirm this model, the tariff values and whether couriers should be allowed to counter-offer.

### What “price locked” does and does not mean

After acceptance neither courier nor platform can change the amount. One exception exists by design: if operations reassign an order before pick-up (for example, an unreachable courier), it returns to searching at the sender’s last offered price, unlocked until another courier accepts; staff do not change the amount, but the sender may raise it again. The sender is notified, not asked for approval. We recommend the customer promise “the price you agreed cannot be changed by the courier”, not an unconditional guarantee.

## 2.4 Matching and assignment

Matching is automatic and sequential. An eligible courier is online, approved, has the right vehicle, a recent position inside the radius, no active job or open offer, room under the cash limit, and is not linked to the sender's account, which blocks fake orders and ratings. The database decides acceptance, so two couriers can never win the same order.

| PARAMETER | STARTING VALUE | NATURE |
|---|---|---|
| Time for one courier to answer an offer | 20 s | |
| Initial search radius | 3 km | |
| Widened radius | 8 km after 60 s | RECOMMENDED Configuration values chosen by the development team, adjustable without code changes, to be tuned with pilot data. |
| Search window before the order expires | 10 min | |
| Courier position considered fresh | 120 s | |
| Vehicle by parcel size | S, M: motorbike · L: car | |
| Active jobs per courier | 1 | BUILT & TESTED Enforced in the database. Batching several parcels per courier is a possible later extension. |

## 2.5 Payments and cash

At launch all money changes hands in cash; no card payment is processed by the platform. Two different amounts are involved and must be kept apart:

- **Cash on delivery for goods**, collected by the courier on behalf of a seller. This is recorded in each courier's append-only ledger, limited by a cash ceiling set at approval, and reconciled through numbered handover receipts entered by operations. BUILT & TESTED
- **The delivery fee**, paid by the sender or the recipient as chosen at ordering, and collected by the courier with the goods amount when the recipient pays. The app makes the courier confirm the total collected. How the fee is shared between courier and platform is not yet defined. CLIENT DECISION

> **COMMERCIAL RULES NOT YET DEFINED**
>
> Commission, courier earnings and payouts, fee accounting, cancellation and waiting fees, return pricing (currently zero), refunds and seller remittance are business decisions, deliberately not invented here. The ledger can absorb them without restructuring; they must be agreed before the pilot.

## 2.6 Launch scope and later extensions

### Proposed launch scope

- Immediate deliveries inside open zones of Algiers
- Small, medium and large parcels; motorbike, car
- Cash on delivery for goods, with courier limits
- Ratings in both directions; problem reports
- Arabic, French, English; Android, then iPhone
- Push notifications; SMS only for the login code

### Deliberately left for later

- Card payment (CIB, Edahabia); courier payouts
- Seller business accounts, bulk orders, API
- Scheduled deliveries; multi-parcel routes
- Deliveries between wilayas
- In-app chat; saved addresses; recipient web page
- Excluded: passengers, food, dangerous goods

SECTION 3

# Prototype and development progress

The prototype is a functioning system, not a set of mock-ups: the screens below are produced by the application's own code, talking to the real backend and database. It has been verified by automated tests, not yet by real couriers on real streets.

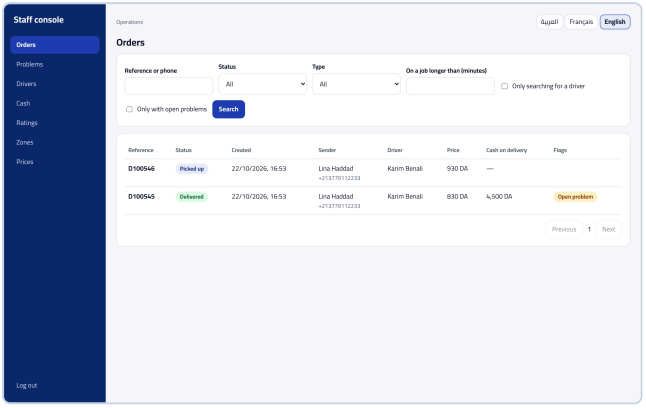
## 3.1 What has been built

| COMPONENT | WHAT WORKS TODAY | EVIDENCE |
|---|---|---|
| **Backend** | Accounts and the two-profile model; one-time-code login with rotating sessions; quotes and versioned pricing; zones; order creation with duplicate protection; sequential matching; offers; job steps; delivery codes; cash ledger and handovers; ratings; problem reports; returns; live notifications with replay; phone-push dispatch; courier document intake; all staff operations. | 56 server functions. 225 functional tests and 92 attack tests, run against a real PostgreSQL database; 12 direct checks of database rules. |
| **Mobile app** | 17 screens covering the complete sender journey and courier journey, settings, profile switching, courier sign-up with photos, incoming parcels for registered recipients, push registration, and offline queuing of courier steps. Arabic, French, English with right-to-left layout. | 28 screen-level tests that drive the real screens against the real backend, including a full order with a courier and a simulated network loss. |
| **Operations console** | Orders and timelines, interventions, problems, couriers and document photos, cash, ratings, zones and pricing; two-factor staff login. | 13 end-to-end tests against the real backend; automatic check that every console call matches a server function. |
| **Android build** | Test build (APK) produced by Expo's cloud build service, with over-the-air update channel configured. | Build completed; not connected to a reachable server. |

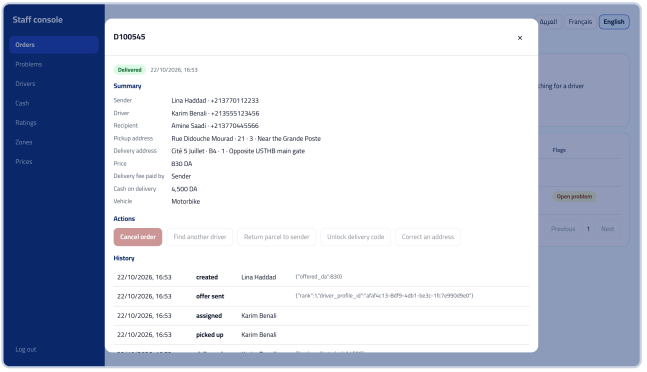
All 358 automated tests and 12 database checks passed in a full run on 22 October 2026. Tests prove that the specified behaviour works as designed; they do not replace device testing, load testing or an independent security assessment, which are planned in Section 6.

## 3.2 The application today

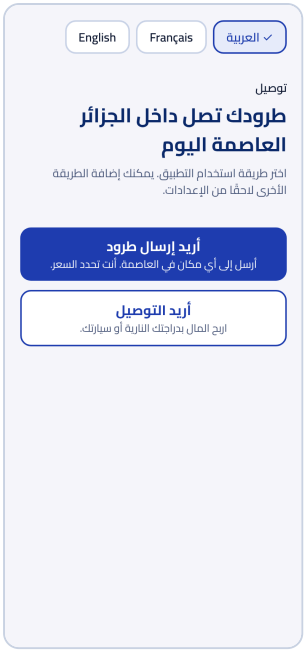
The operations console and the mobile app, as they run today.



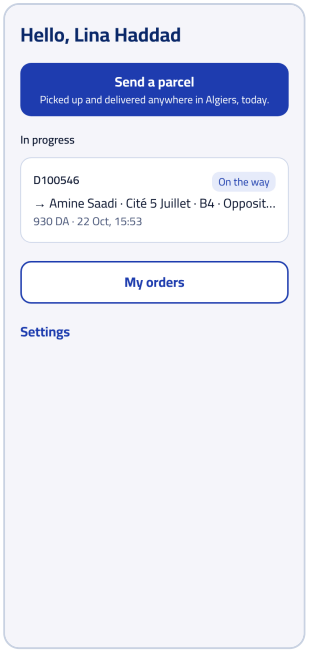

**Console — orders.** Filters by status, type, age, open problems; reference or phone search.



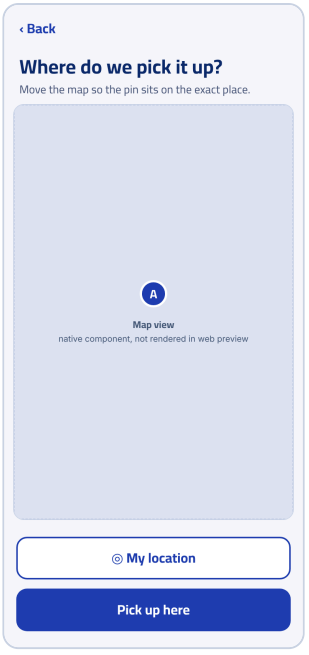

**Console — order detail.** Parties, addresses, amounts, permitted actions and full history. The history currently shows some raw technical data; formatting it is listed as remaining work.



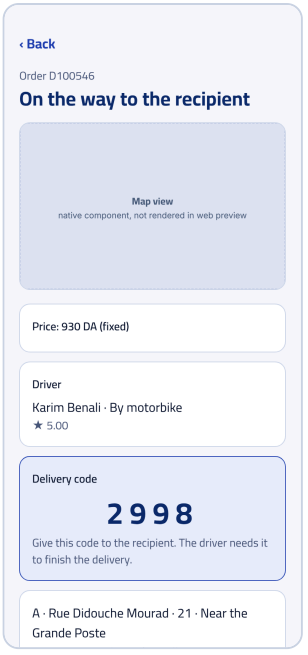

**Welcome, Arabic.** Path choice and instant language switch.



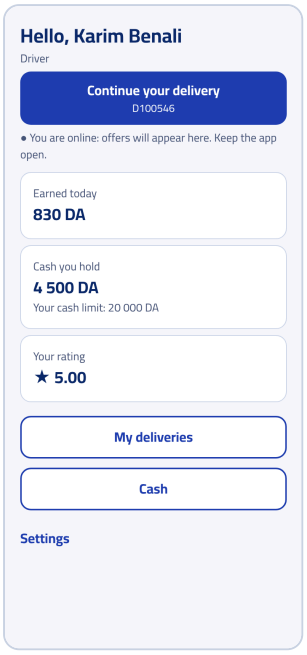

**Sender home.** Orders in progress and quick access.



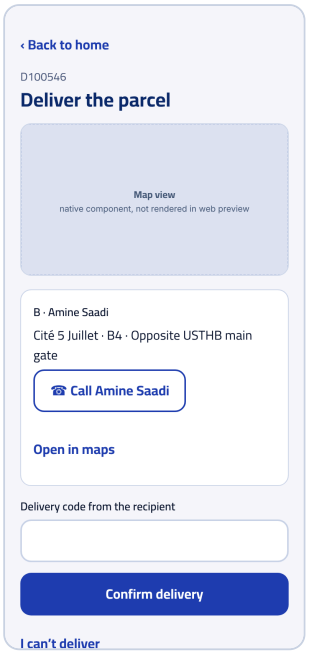

**Pick-up point.** Pin under a fixed centre; map placeholder in this preview.



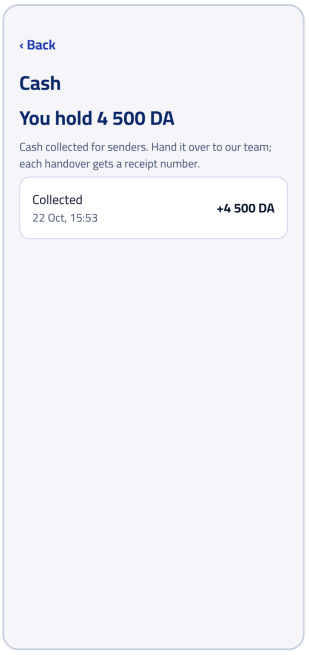

**Live order.** Fixed price, courier, delivery code.



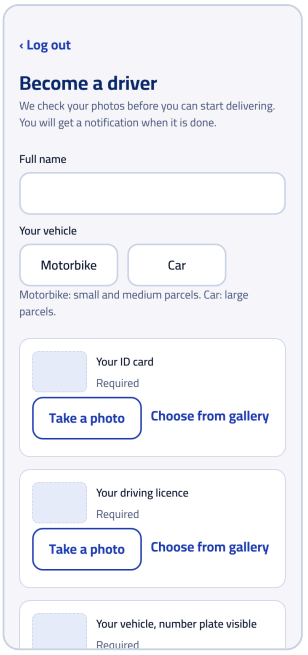

**Courier home.** Current job, earnings shown as order value, cash held against limit.




**Active job.** One action at a time; call, navigate, enter code.




**Courier cash.** Collected amounts; handovers receive receipts.




**Courier sign-up.** Vehicle and three document photos.

Screens captured on 22 October 2026 from the application's code running in a web preview, connected to the test backend with fictitious test data (names, numbers, prices). The map is a native component that cannot run in a web preview, so it appears as a labelled placeholder; on the phone it is an interactive map. Visual details on Android may differ slightly.

## 3.3 What is simulated, incomplete or unverified

The following items separate a tested prototype from a production service. None requires redesign, but each needs engineering, an external account, or verification.

| AREA | CURRENT STATE | NEEDED FOR PRODUCTION |
|---|---|---|
| **Hosting** | Runs only in the development environment. Deployment scripts for a single server exist but are untested. | Production and staging environments (Section 5.4). |
| **SMS login codes** | Simulated: codes are generated, limited and verified for real, but not sent by SMS. | Select an SMS gateway with Algerian coverage; integrate and test delivery rates. |
| **Phone push** | Wired in the server and app; never tested with a live Firebase project. | Firebase project owned by the client; test on Android and iPhone. |
| **Maps and routing** | Open-source map (MapLibre with OpenStreetMap tiles, confirmed to load real Algiers streets). Road-distance engine tested only on a synthetic grid. Straight-line estimate used as fallback. | Replace with Google Maps Platform for map, address search and travel times (Section 5.2). |
| **Courier location** | Sent only while the app is on screen. Tracking pauses when the courier switches to a navigation app. | Location service during active jobs (Section 5.3). |
| **Document photos** | Stored on the server's local disk. | Move to private cloud storage; managed containers have no permanent disk. |
| **Scaling details** | Rate limits held in one server's memory; notification history has no retention period. | Shared limits in Redis before running two or more servers; retention policy. |
| **Mobile releases** | Android test build only; iPhone build not produced; over-the-air updates not yet signed. | Store accounts, iPhone build, update signing, store listings. |
| **Verification** | No field test, device matrix, load test, external penetration test or native-speaker review yet. | Phase 3 of the roadmap. |
| **Commercial logic** | No commission, payout, fee or refund logic, by design (Section 2.5). | Client decisions, then implementation. |

SECTION 4

# Solution architecture

The architecture keeps every business rule on the server, where it cannot be bypassed from a phone, and keeps the rules that protect money inside the database, where a fault in application code cannot override them. Around that core sit three client applications and a small number of external services.

## 4.1 System context

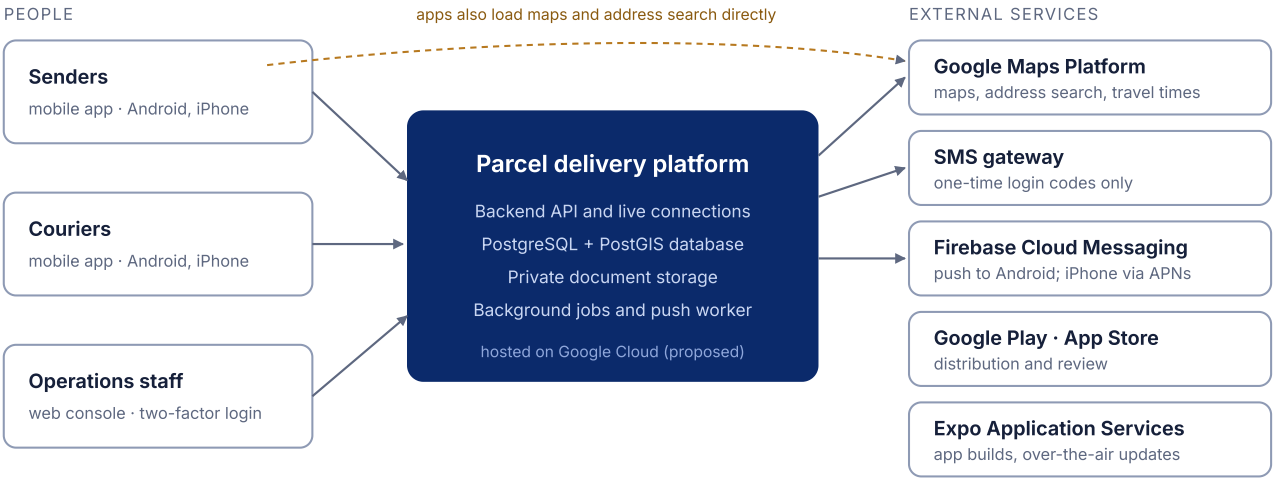



**Figure 4 — System context.** All business operations go through the platform. The apps call Google Maps directly only for map display and address search; prices and travel times are calculated on the server.

## 4.2 Data and integrity

PostgreSQL is the single source of truth. The schema is created only by versioned migrations and production starts empty: no sample data. The application connects with a restricted database role, so the rules below hold even if application code is faulty or compromised.

| RULE | HOW IT IS ENFORCED | CONTROL |
|---|---|---|
| **Only valid order transitions** | Database trigger rejects any other status change; final states are immutable. | Prevent |
| **Price fixed at acceptance** | Database rejects changes to an assigned order's price. | Prevent |
| **One courier wins an order; one active job per courier** | Atomic conditional update; unique index on active jobs. | Prevent |
| **Cash records cannot be edited or deleted by the application** | Ledger is append-only for the application role; corrections are new entries; a reversal can occur only once. | Prevent, trace |
| **History, ratings, staff audit and security log are permanent** | No update or delete rights for the application role. | Trace |
| **Suspicious activity is visible** | Security event log (token reuse, privilege probing, rate-limit hits, wrong codes); daily report. | Detect |

These controls bind the application. A database administrator with full privileges could still alter data; in production that access is restricted to named individuals, protected by two-factor login and recorded in the cloud provider's audit logs, and backups allow comparison and recovery. We therefore describe the ledger as append-only and auditable rather than tamper-proof.

## 4.3 Application and backend architecture

The backend is one deployable application divided into modules with clear responsibilities. This *modular monolith* is faster to build, cheaper to run and simpler to secure than separate microservices, while keeping boundaries that allow a module to be extracted later if its load justifies it.

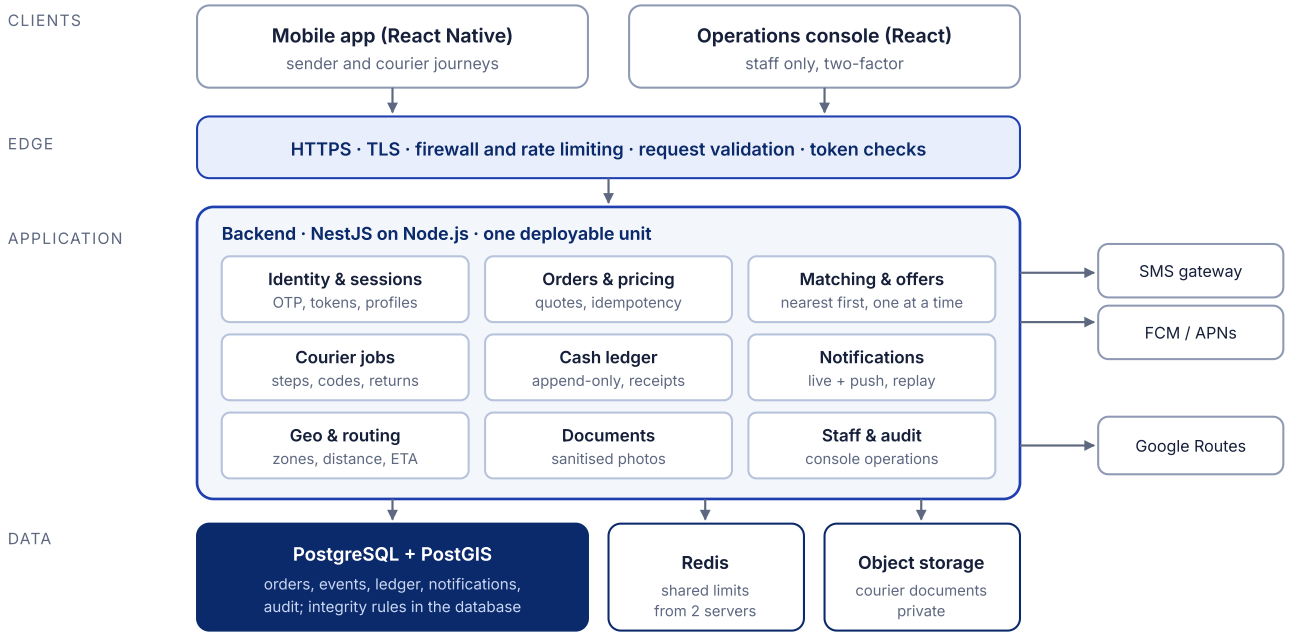



**Figure 5 — Logical architecture.** Each module owns its part of the data and is reached through defined interfaces. Redis is added when two or more backend instances run. Security middleware, scheduled maintenance and the push worker are part of the same deployable unit.

## 4.4 Security architecture

No system can honestly be called completely secure. The objective is defence in depth: independent layers, so that defeating one does not open the others; a trace for every attempt; and no single secret that unlocks everything.

| LAYER | CONTROLS IN THE PROTOTYPE | ADDED FOR PRODUCTION |
|---|---|---|
| **Customer login** | One-time SMS codes stored only as keyed hashes, 5 guesses per code counted atomically; 15-minute access tokens; refresh tokens rotated on each use, with reuse treated as theft; maximum 5 devices; new-device alerts; log out everywhere. | Real SMS provider; anti-fraud caps already limit SMS cost abuse. |
| **Staff login** | Password plus authenticator code; identical error for any wrong factor; lockout; 2-hour sessions; encrypted authenticator secrets. | Named accounts only; periodic access review. |
| **Requests** | Strict validation of every field, size limits, security headers, per-address and per-session rate limits, object-level access checks; fuzz-tested. | Web application firewall and DDoS filtering at the edge; shared limits across servers. |
| **Data** | Restricted database role; integrity rules in the database; documents re-encoded to remove hidden data, private and viewable only by staff, each view audited. | Encryption at rest, private networking, secrets in a managed vault. |

| LAYER | CONTROLS IN THE PROTOTYPE | ADDED FOR PRODUCTION |
|---|---|---|
| **Mobile** | Session key in the phone's secure keystore; release builds refuse non-HTTPS servers; one token renewal at a time. | Signed over-the-air updates; Play Integrity and App Attest checks. |
| **Assurance** | An automated attack test for every server function; dependency audits. | Independent penetration test before launch; incident and breach-response procedure. |

## 4.5 Key flow: from order to confirmed delivery

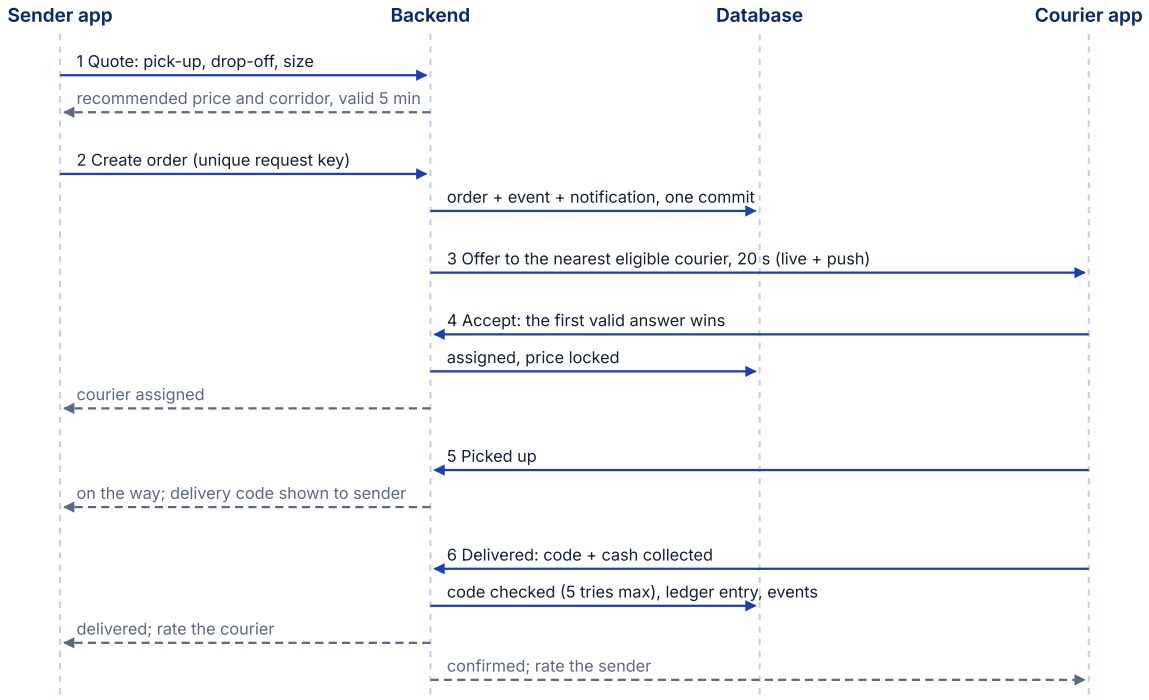



**Figure 6 — Order sequence.** Solid arrows are requests and database writes; dashed arrows are notifications. Every state change and its notifications are written in a single database transaction. Steps 5 and 6 can be queued on the phone when there is no network; the delivery is complete only when the server accepts the code.

### When is a delivery officially complete?

Only when the server has accepted the correct delivery code and recorded the delivery. If the courier has no network at the door, the app saves "picked up" or "delivered" on the phone, tells the courier that the step will be sent automatically, and retries until the server answers. Repeating a step is harmless, so a retry never duplicates anything. If the server later refuses a queued step, for example because the code was wrong, the courier is told immediately and asked to contact support.

RECOMMENDED The weak point is a wrong code entered offline: the courier may already have left the recipient. We do not recommend checking the code on the phone, because four digits stored on the phone could be recovered and misused. We recommend instead that the app shows such deliveries as "awaiting confirmation" and asks the courier to stay until confirmed where coverage allows, and that operations receive an alert for any delivery still unconfirmed after a set time.

## 4.6 Real-time updates and notifications

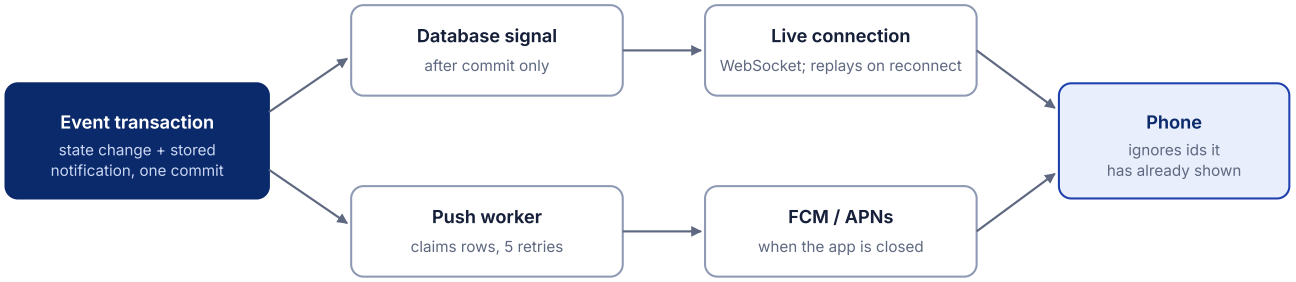



**Figure 7 — Notification delivery.** Notifications are stored durably with the event that causes them, then delivered over the live connection and, when the app is closed, by phone push. After a reconnection the app sends the last identifier it saw and receives, in order, everything since that is still valid.

### What the design guarantees

- No notification without its event, and no event without its notification: both are committed together.
- No duplicates: one notification per person and event, and the app ignores identifiers it has already shown.
- Nothing still relevant is lost while a phone is offline: it is replayed on reconnection.
- Each profile receives only its own notifications.

### What it cannot guarantee

- Immediate arrival on a device that is offline, in battery saving mode, or restricted by the phone's operating system.
- Delivery of phone push, which depends on Google and Apple services; it is retried up to five times.
- Replay of expired items: an offer that has lapsed is not shown again.
- A retention period for stored notifications has still to be set.

SECTION 5

# Technology and infrastructure strategy

Every component below is mainstream, well supported and replaceable. Where the prototype already uses a technology and it remains appropriate, we keep it. Where production needs something different, mapping and hosting above all, we say so and explain why.

## 5.1 Technology choices

| LAYER | TECHNOLOGY | WHY IT FITS | TRADE-OFF | STATUS |
|---|---|---|---|---|
| **Language** | TypeScript, end to end | One language and type system for app, server and console; errors caught before release; one team covers every layer. | Requires discipline on shared types and versions. | BUILT |
| **Mobile app** | React Native 0.88 with Expo SDK 58 | One codebase for Android and iPhone; cloud builds, including iPhone builds without a Mac; over-the-air delivery of fixes. | Native features need development builds; follows Expo's release cycle. | BUILT |
| **Backend** | NestJS on Node.js 22 LTS | Structured modules and guards suited to a modular monolith; efficient for many concurrent connections. | CPU-heavy work would need separate workers; none is planned. | BUILT |
| **Database** | PostgreSQL 16 with PostGIS | Transactions, constraints and triggers enforce money rules; PostGIS finds nearby couriers and checks zones. | Single primary for writes; ample for city-scale volumes. | BUILT |
| **Real time** | WebSocket with database notifications | Instant updates with replay, without a separate message broker. | Connections must be re-established after server restarts; built in. | BUILT |
| **Shared cache** | Redis (managed) | Shared rate limits once two or more servers run. | An extra managed service to pay for and monitor. | RECOMMENDED |
| **Maps and routing** | Google Maps Platform | Familiar maps, address and place search, traffic-aware travel times (Section 5.2). | Per-request costs above free allowances; strict terms. | RECOMMENDED |
| **Push** | Firebase Cloud Messaging | Standard route to Android, and to iPhone through Apple's service; no charge for messaging. | Delivery timing controlled by Google and Apple. | PARTLY BUILT |
| **Hosting** | Google Cloud: Cloud Run, Cloud SQL | Managed containers and database; no servers to patch; scales in steps (Section 5.4). | No region in Algeria; legal review required (Section 5.6). | RECOMMENDED |
| **SMS** | Licensed gateway with Algerian coverage | Login codes only; anti-fraud caps already built. | Main variable cost per new login. | CLIENT DECISION |

## 5.2 Mapping: from the prototype's open stack to Google Maps

The prototype uses open-source mapping: MapLibre with OpenStreetMap vector maps, which we confirmed load real Algiers streets, and a self-hosted routing engine with a straight-line fallback. That choice avoided costs during development. For production we recommend Google Maps Platform, mainly for address and place search, which the open stack cannot provide at application scale without significant extra work, and for traffic-aware travel times that make prices and arrival estimates more credible.

| | OPEN STACK (PROTOTYPE) | GOOGLE MAPS PLATFORM (PROPOSED) |
|---|---|---|
| **Address entry** | Map pin plus typed street, building, floor and landmark | Search by place, street or landmark, then confirm on the map; the typed details are kept |
| **Travel time and distance** | Road network without live traffic; must be hosted and refreshed | Traffic-aware routes, computed on the server for each quote |
| **Operating effort** | Map and routing servers to run and update | None; quotas and budget alerts to manage |
| **Cost model** | No per-request fees; hosting effort instead | Free monthly allowance per service, then per-request pricing (Section 7.1) |
| **Terms** | Attribution required | No use of Google data on non-Google maps; limited caching; restrictions on some uses of place coordinates |

### Consequences for the design

- **All or nothing.** Google's terms prohibit using its content with or near a non-Google map. Google data therefore cannot be added on top of the prototype's open-source map: the map component itself must be replaced. On iPhone this rules out Apple Maps as the display, so we recommend the react-native-maps library with the Google provider on both platforms; its current release supports React Native's new architecture. RECOMMENDED
- **Server-side routing.** Prices and arrival times are calculated on the server through the Routes API, so the price logic cannot be manipulated from the phone. Route results may not be stored beyond what the terms allow, so each quote makes its own request, protected by a timeout and the existing straight-line fallback.
- **Zone checks.** Service zones are checked against the position the sender confirms on the map. Google's terms restrict using Places coordinates for point-in-polygon analysis; the precise treatment of a pin placed from a search result should be confirmed against the terms before launch. TO BE VERIFIED
- **Cost control.** Address-search sessions, server-side quotas per service, and budget alerts on the billing account keep spending predictable.

## 5.3 Courier location during deliveries

The prototype sends the courier's position only while the app is on screen, every 15 seconds or 25 metres. That is the most privacy-friendly option and avoids sensitive store reviews, but it has a practical flaw: when the courier opens Google Maps to navigate, the app goes to the background and the sender's tracking freezes until the courier returns to it.

| OPTION | EFFECT | REQUIREMENTS |
|---|---|---|
| **Keep foreground only (current)** | Tracking pauses during navigation; positions resume when the app is reopened. | None. |
| **Location service while online and on a job** RECOMMENDED | Tracking continues during navigation. A persistent notification tells the courier that location is being shared; it stops when the courier goes offline. | Android: a foreground service of type “location” started while the app is visible, which does not need the background-location permission; a Play Console declaration with a demonstration video. iPhone: background location mode with a visible system indicator. |
| **Always-on background location** | Continuous tracking even when offline. | Background-location permission and stricter store review. Not recommended: more intrusive than the service needs. |

Android behaviour is confirmed in Android’s developer documentation and supported by the app’s location module; iPhone behaviour must be verified on a device, and Google Play’s announced “minimum scope” location policy reviewed before submission. CLIENT DECISION Confirm the option, with a short privacy notice to couriers.

## 5.4 Hosting and deployment

Google Cloud has no region in Algeria or North Africa. The nearest regions are Madrid, Milan, Turin and Paris; latency from Algiers should be measured before choosing. The backend is packaged as a standard container, so it can run on any of the options below, or with a provider in Algeria if the legal review requires it.

| OPTION | CHARACTERISTICS | ASSESSMENT |
|---|---|---|
| **A. Single virtual server** | One machine running the backend, database and web server. Deployment scripts already exist for this model but are untested. | Lowest cost; single point of failure; more operational work. Fallback if hosting must be local. |
| **B. Lean managed** PILOT | Cloud Run (one always-on instance) with Cloud SQL for PostgreSQL in one zone, automated backups and point-in-time recovery; Cloud Storage for documents; Secret Manager. | Little operational work, predictable cost, recovery from backups. Right for a closed pilot. |
| **C. High availability** LAUNCH | Option B plus a standby database in a second zone with automatic failover, two or more backend instances, managed Redis, a load balancer and Cloud Armor firewall, and a separate staging environment. | Survives the loss of a server or a zone with a short interruption. Recommended before public launch. |
| **D. Kubernetes (GKE)** | Container orchestration with full control of networking and workloads. | Not justified now: a cluster management fee and more operational skill for no current benefit. |

### Compatibility of the existing code with this design

- Cloud Run supports WebSockets, but limits a single connection to at most 60 minutes. The app already reconnects automatically and replays anything missed, so this limit is absorbed. BUILT
- Background jobs need instance-based billing and at least one always-on instance. With several instances, matching already runs under a database lock and the push worker claims work row by row, so jobs are not duplicated. BUILT
- Rate limits must move to Redis, and document photos to Cloud Storage, before a second instance runs. REMAINING WORK

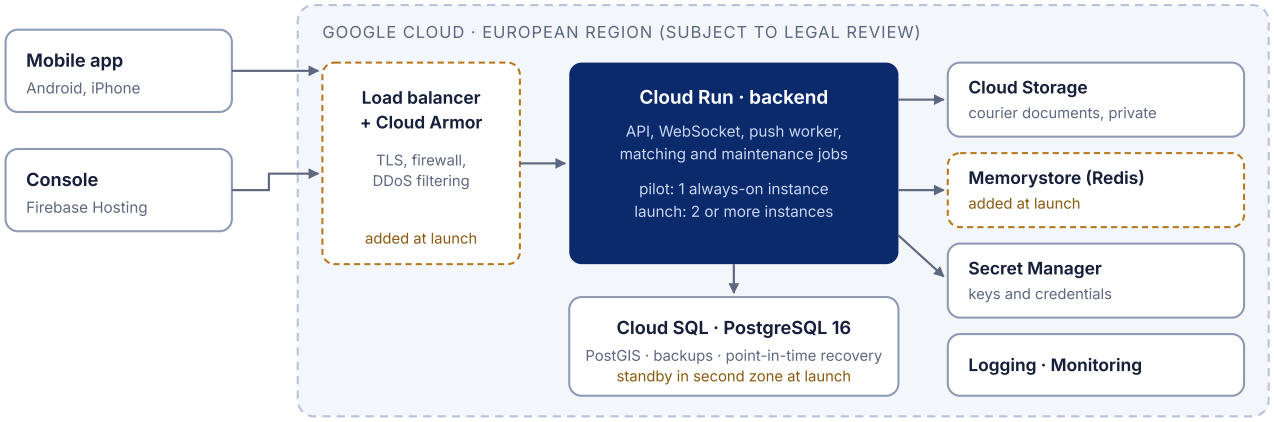



**Figure 8 — Recommended deployment.** Solid outlines: the lean pilot configuration (option B). Amber dashed outlines: components added for public launch (option C). Google Maps, the SMS gateway and Firebase Cloud Messaging are reached from the backend over HTTPS.

## 5.5 Operations: environments, releases, backups, monitoring

### Environments and releases

- Separate staging and production; every change runs the full test suite, and a failure blocks it.
- Backend releases as new Cloud Run revisions, with gradual traffic shift and one-step rollback; database migrations applied first and backwards-compatible.
- Mobile: small fixes by signed over-the-air updates; native changes through store releases with staged rollout.

### Backups and monitoring

- Daily backups and point-in-time recovery, copied to a second region; documents versioned; a recorded restore test every month.
- Uptime checks, error reporting, structured logs and dashboards; crash reports from both apps.
- Alerts for failures, slow matching, unconfirmed deliveries and unusual security events; a daily security report.

## 5.6 Reliability targets and data protection

**Targets, not commitments.** Service targets should be agreed after the pilot, from measured behaviour. As a starting point for public launch we propose: 99.9% monthly availability of ordering and delivery; data loss limited to minutes in a database failure (point-in-time recovery); recovery from a zone failure within minutes with the standby database, or within hours from backups in the lean configuration. These describe design intent, not tested results or contractual service levels.

**Data protection.** The platform processes phone numbers, names, addresses, courier locations and identity documents. Algeria's Law 18-07 (reported to have been amended by Law 25-11 of July 2025) requires prior declaration to, or authorisation by, the national authority (ANPDP); secondary sources indicate that transferring personal data abroad, which may include hosting in a European region, needs its prior authorisation, and Law 18-05 on e-commerce requires some online commercial services to be hosted in Algeria. TO BE VERIFIED We recommend a written opinion from Algerian counsel on these points before the hosting region is chosen. The design does not depend on the answer: if local hosting is required, option A runs with an Algerian provider.

SECTION 6

# Development roadmap

The path from prototype to production runs through four engineering phases and one business track. Each phase ends with explicit completion criteria. Durations are not stated yet because they depend on decisions and lead times listed at the end of this section; we will commit to a schedule once those are known.

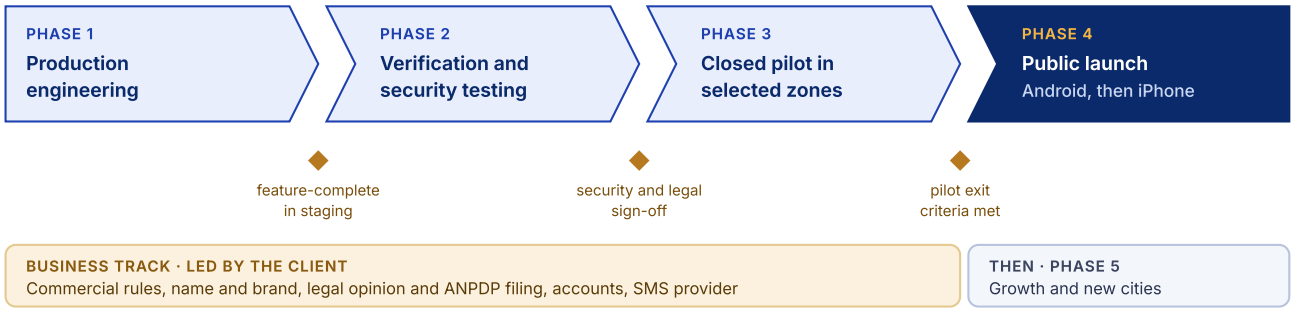



**Figure 9 — Roadmap.** Phases are shown in sequence, not to scale. The business track runs in parallel and gates phase 3: the pilot needs agreed commercial rules, legal sign-off and live accounts.

| PHASE | MAIN DELIVERABLES | COMPLETION CRITERIA |
|---|---|---|
| **1. Production engineering** | Google Maps integration (map, address search, server-side routes); courier location service; SMS gateway; live push through the client's Firebase project; document storage in the cloud; shared rate limits; notification retention; signed over-the-air updates; readable order history in the console; commercial logic once decided; staging and production environments defined as code; release pipeline and monitoring; iPhone build; current Google Play target API level. | All functions complete in staging; full test suite passing, with attack tests for every new function; no open high-severity defects. |
| **2. Verification** | Testing on a range of Android phones, including low-cost models; end-to-end runs on staging; load test of matching and live connections; independent penetration test; native-speaker review of Arabic and French; terms of service and privacy notices; store listings and declarations; ANPDP filing. | No unresolved critical or high security findings; load targets met; legal sign-off on hosting and data processing; store declarations accepted. |
| **3. Closed pilot** | Selected Algiers zones; a recruited group of couriers and sellers; Google Play closed-testing track; operations playbook; daily review; tuning of matching and price parameters. | Exit targets agreed with the client before the pilot (for example matching time, completion rate, cash reconciliation and crash-free sessions) met over the agreed period. |
| **4. Public launch** | High-availability configuration (option C); public Android release with staged rollout; iPhone release after App Store review; zones opened progressively. | Production-readiness review passed: restore tested, alerts and on-call in place, support procedures ready. |
| **5. Growth** | Candidate extensions: card payment (CIB, Edahabia), seller accounts and API, scheduled deliveries, multi-parcel routes, new cities. | Prioritised from pilot and launch data. |

## 6.1 What is needed to commit to a schedule

- **Team and availability.** The size and allocation of the engineering team for phases 1 and 2.
- **Decisions.** The commercial rules and the pricing model in Section 2; changes there affect phase 1 scope.
- **Legal opinion.** Hosting location and ANPDP procedure, whose processing time is not published.
- **Account lead times.** Organisation accounts for Google Play and Apple require a D-U-N-S number, which can take from several business days to several weeks.
- **Third parties.** SMS provider onboarding, penetration-test availability, app store review.
- **Pilot design.** Number of couriers and zones, and the minimum pilot duration the client wants.

## 6.2 Resources and ongoing operation

| ROLE | INVOLVEMENT |
|---|---|
| **Full-stack and mobile engineering** | Phases 1–4, then maintenance: security updates, operating-system and store requirement changes, which recur every year. |
| **Quality assurance** | Device testing in phases 2–3 with a set of real Android phones and at least one iPhone. |
| **Operations staff** | From the pilot: courier approval, live monitoring, problem handling, cash reconciliation. |
| **External specialists** | Penetration tester; Algerian legal counsel; native-speaker reviewers for Arabic and French. |

SECTION 7

# Costs, risks and recommendations

Running costs at pilot scale are modest and largely predictable; the main variables are mapping requests, SMS volume and the size of the database. The main risks are not technical: they are courier supply, cash handling, regulation and the commercial rules still to be decided.

## 7.1 Cost categories

Figures are list prices published by the providers, in US dollars, before tax and discounts, and assume 730 hours per month. Cloud SQL, Memorystore and load-balancing prices were read for the provider's default region (Iowa, United States); European regions may differ and must be checked in Google's pricing calculator. Cloud Run prices apply equally to Madrid and Paris. No development cost is given: it depends on the team composition, which has not been defined.

| CATEGORY | BASIS | INDICATIVE COST | SOURCE STATUS |
|---|---|---|---|
| **Hosting — pilot (option B)** | Cloud Run, one always-on instance (1 vCPU, 1 GiB) ≈ US$47 after free allowance; Cloud SQL Enterprise, 1 vCPU / 4 GiB, single zone ≈ US$51, plus storage at ≈ US$0.17 per GB-month; small amounts for document storage, secrets and logs | ≈ US$100–130 / month | Pricing pages read |
| **Hosting — launch (option C)** | Two backend instances; high-availability database at twice the single-zone rate (≈ US$101 for 1 vCPU / 4 GiB, ≈ US$202 for 2 vCPU / 8 GiB); Redis 1 GiB standard ≈ US$47; load balancer ≈ US$18; Cloud Armor ≈ US$5 per policy, US$1 per rule, US$0.75 per million requests | ≈ US$280–380 / month | Pricing pages read |
| **Google Maps Platform** | Free monthly allowance per service, then per 1,000 requests. Indicative list prices: place details (essentials) US$5; routes US$5, or US$10 with live traffic; autocomplete requests US$2.83. Map display in the mobile apps reported free without a custom map style | see estimate below | TO BE VERIFIED |
| **SMS login codes** | Per message, by provider; volume follows new logins and new devices, not orders, because sessions persist | quote required | — |
| **Push and crash reporting** | Firebase Cloud Messaging and Crashlytics | no charge | Reported by provider |
| **App distribution** | Google Play, one-time registration; Apple Developer Program, yearly (Apple notes prices may vary by region) | US$25 once · US$99 / year | Apple page read; Google reported |
| **Security testing** | Independent penetration test before launch; repeated after major changes | quote required | — |
| **Legal and language** | Algerian counsel (data protection, e-commerce, terms); native-speaker review | quote required | — |

| CATEGORY | BASIS | INDICATIVE COST | SOURCE STATUS |
|---|---|---|---|
| **Maintenance** | Security updates, yearly store and operating-system requirements, monitoring and support | team time | — |

**ILLUSTRATIVE MAPPING COST PER ORDER**

Assumptions: two address searches per order, each completed by one place-details request; on average 1.5 traffic-aware route requests per order (some senders change the quote); map display free. On these assumptions the free monthly allowances (reported as 10,000 place-details and 5,000 traffic-aware route requests) would cover roughly 3,000 orders per month. Beyond that, the indicative cost is about US$0.025 per order (2 × US$0.005 + 1.5 × US$0.010). These prices could not be confirmed on Google's own price list for this proposal and must be checked before budgeting.

## 7.2 Principal risks

| RISK | LIKELIHOOD | IMPACT | MITIGATION |
|---|---|---|---|
| **Too few couriers at launch** | High | High | Pilot in a few zones; recruit couriers before opening a zone; zones opened only as supply allows; senders can raise the offer. |
| **Cash-on-delivery losses** | Medium | High | Per-courier cash limits, append-only ledger, numbered receipts, document checks at approval, a daily reconciliation routine. |
| **Regulatory constraints on hosting and data** | Medium | High | Counsel's opinion before choosing the region; ANPDP filing in the business track; portable design allowing local hosting. |
| **Commercial rules decided late** | Medium | High | Decide in the business track; the pilot cannot start without them. |
| **Fraud and fake accounts** | Medium | Medium | Same-account block, phone verification, document review, one rating per order, SMS caps, audit trail. |
| **Delivery disputes** | Medium | Medium | Server-confirmed delivery codes, automatic returns, problem desk, alerts on unconfirmed deliveries. |
| **Weak networks and low-cost phones** | High | Medium | Offline step queue, polling fallback, device testing on low-cost Android models. |
| **Store review of location features** | Medium | Medium | Minimal location scope, clear disclosure, Play Console declaration and demonstration video. |
| **Mapping cost growth or terms breach** | Medium | Medium | Quotas and budget alerts; Google-only map display; design reviewed against the terms. |
| **Security incident** | Low | High | Layered controls, penetration test, monitoring, an incident and breach-notification procedure aligned with the law. |

Likelihood and impact are qualitative judgements by the development team, to be revisited after the pilot.

## 7.3 Decisions required from the client

| # | DECISION | OUR RECOMMENDATION |
|---|---|---|
| 1 | **Pricing model** | Guided price with corridor and raise-while-searching; no courier counter-offers; price locked at acceptance with the reassignment rule in Section 2.3. |
| 2 | **Commission, courier earnings and fees** | Define before the pilot: commission, payout method, delivery-fee settlement, cancellation, waiting and return fees, refunds. |
| 3 | **Courier location** | Location service while online and on a job, with a persistent notice (Section 5.3). |
| 4 | **Hosting location** | Decide after counsel's opinion; Google Cloud European region if permitted, otherwise local hosting. |
| 5 | **SMS provider** | Select after quotes, comparing delivery rates on Algerian networks. |
| 6 | **Name, brand and accounts** | Final name and Android package name before the first store upload; Google Cloud, Firebase, Play and Apple accounts in the company's name. |
| 7 | **Pilot design** | Zones, number of couriers and sellers, duration and exit targets. |
| 8 | **Liability and insurance** | Responsibility for lost or damaged goods; declared-value limits; list of prohibited goods. |

## 7.4 Recommendations and next steps

1. **Confirm the operating model** in Section 2, or tell us what should change, so that phase 1 scope can be fixed.
2. **Start the business track now:** appoint Algerian counsel, register the company accounts (beginning with the D-U-N-S number, which has the longest lead time), and request SMS quotes.
3. **Approve phase 1** and the engineering team; we will then issue a schedule and a budget for development, based on the information listed in Section 6.1.
4. **Approve Google Maps and the location option**, with a monthly spending cap and alerts on the billing account.
5. **Agree the pilot**: zones, participants, duration and the measurable exit targets.

The prototype already demonstrates the core of the service: guided pricing, automatic matching, protected price, proof of delivery and accountable cash. The work ahead is to connect it to production services, verify it independently and prove it with real couriers, in that order.

APPENDIX A

# Sources, assumptions and glossary

## Verification of external facts

External facts were checked on 22 October 2026. “Read” means the source page itself was consulted. “Reported” means only a search summary of the official page was available; such facts must be confirmed before they are relied on contractually.

| TOPIC | SOURCE | STATUS |
|---|---|---|
| **Cloud Run prices and region tiers** | cloud.google.com/run/pricing | Read |
| **Cloud SQL prices; HA at twice zonal rate** | cloud.google.com/sql/pricing (Iowa prices) | Read |
| **GKE management fee and free credit** | cloud.google.com/kubernetes-engine/pricing | Read |
| **Memorystore, Cloud Armor, load balancing prices** | cloud.google.com pricing pages for each service | Read |
| **Google Cloud regions; none in North Africa** | cloud.google.com/about/locations (updated 13 October 2026) | Read |
| **Cloud Run WebSockets (60-minute limit), billing modes; Cloud SQL PostGIS, HA failover, recovery windows** | Google Cloud documentation pages | Reported |
| **Google Maps Platform terms: no use with non-Google maps, caching limits, place-coordinate restrictions; prohibited territories (Algeria not listed)** | cloud.google.com/maps-platform/terms and service-specific terms (October 2026) | Read |
| **Google Maps Platform prices and free allowances** | developers.google.com/maps/billing-and-pricing | Reported |
| **react-native-maps 1.29.x and new-architecture support; expo-maps stable in SDK 58** | Published npm packages and change logs | Read |
| **Android foreground location service without background permission; Play target API 36 from 31 August 2026** | developer.android.com (updated October 2026) | Read |
| **Play Console declarations; personal-account testing rule; D-U-N-S for organisations; US$25 fee** | support.google.com (Play Console Help) | Reported |
| **iPhone background location with when-in-use permission** | developer.apple.com (Core Location documentation) | Read |
| **Apple Developer Program fee and organisation enrolment** | developer.apple.com/programs/enroll | Read |

| TOPIC | SOURCE | STATUS |
|---|---|---|
| **Law 18-07; Law 25-11 (July 2025); cross-border transfer authorisation; Law 18-05 hosting rule** | Law-firm alerts, academic and press summaries | Reported |
| **Firebase Cloud Messaging and Crashlytics at no charge** | firebase.google.com/pricing | Reported |

## Assumptions

- Launch in Algiers, with immediate deliveries only.
- Cash is the only payment method at launch.
- Pilot volumes of a few thousand orders per month or less.
- Hosting in a Google Cloud European region, subject to legal review.
- Android first; iPhone after App Store review.
- Matching and pricing parameters will be tuned with pilot data.

## Glossary

| | |
|---|---|
| **Modular monolith** | One deployable application divided into independent business modules. |
| **Append-only ledger** | A record to which entries can be added but not changed; corrections are new entries. |
| **Idempotent** | Safe to repeat: sending the same request twice has the effect of sending it once. |
| **WebSocket** | A permanent connection that lets the server push updates to the app instantly. |
| **Over-the-air update** | An app update delivered without a new store download, for non-native changes. |
| **Point-in-time recovery** | Restoring the database to any moment within a retention window. |
| **Penetration test** | An authorised attempt by an independent expert to break in. |
| **ANPDP** | Algeria's national authority for the protection of personal data. |